

LAPORAN PRAKTIKUM 1 - STRUKTUR SISTEM OPERASI



Dosen Pengampu:

Triawan Adi Cahyanto M.Kom

Disusun Oleh:

Nabila Senja Ramadhani (2410651043)

PROGRAM STUDI TEKNIK INFORMATIKA

FAKULTAS TEKNIK

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1. Operating System Services

Praktik 1.1: Menjelajahi Layanan OS

```
Installing, this may take a few minutes...
wsl: Failed to start the systemd user session for 'root'. See journalctl for more details.
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: senjaramadhani
New password:
Retype new password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Oct 14 22:31:49 WIB 2025

System load:  0.2               Processes:      54
Usage of /:   0.1% of 1086.85GB  Users logged in: 0
Memory usage: 14%              IPv4 address for eth0: 172.20.159.185
Swap usage:   0%

This message is shown once a day. To disable it please create the
/home/senjaramadhani/.hushlogin file.
senjaramadhani@LAPTOP-823QCJCF:~$ uname -a
Linux LAPTOP-823QCJCF 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
senjaramadhani@LAPTOP-823QCJCF:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
senjaramadhani@LAPTOP-823QCJCF:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.2 165788 10752 ?        Ss   22:32   0:00 /sbin/init
root         2  0.0  0.0   3072  1792 ?        SL   22:32   0:00 /init
root        61  0.1  0.3 396332 14724 ?        Ss   22:32   0:00 /lib/systemd/systemd-journald
root       86  0.0  0.1 22720  5632 ?        Ss   22:32   0:00 /lib/systemd/systemd-udevd
root     107  0.0  0.3 25672 13360 ?        Ss   22:32   0:00 /lib/systemd/systemd-resolved
systemd+ 126  0.0  0.1 89364  7168 ?        Ssl  22:32   0:00 /lib/systemd/systemd-timesyncd
root     187  0.0  0.0   4308  2688 ?        Ss   22:32   0:00 /usr/sbin/cron -f -p
message+ 189  0.0  0.1   8580  4608 ?        Ss   22:32   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
og-only
root     193  0.0  0.4 30136 19072 ?        Ss   22:32   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
syslog   194  0.0  0.1 222404  4992 ?        Ssl  22:32   0:00 /usr/sbin/rsyslogd -n -iNONE
root     197  0.0  0.1 15328  7424 ?        Ss   22:32   0:00 /lib/systemd/systemd-logind
root    221  0.0  0.0   3240  2176 hvc0     Ss+  22:32   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220
root    226  0.0  0.0   3196  2176 tty1     Ss+  22:32   0:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root    228  0.0  0.5 107164 21760 ?        Ssl  22:32   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
gnal
root    281  0.0  0.0   3076   896 ?        Ss   22:32   0:00 /init
root    282  0.0  0.0   3092  1024 ?        S   22:32   0:00 /init
senjara+ 283  0.0  0.1 6128  4992 pts/0    Ss   22:32   0:00 -bash
root    284  0.0  0.1  7528  4864 pts/1    Ss   22:32   0:00 /bin/login -f
senjaramadhani@LAPTOP-823QCJCF:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.7Gi          362Mi        3.3Gi          3.0Mi        128Mi        3.2Gi
Swap:          1.0Gi           0B          1.0Gi

senjaramadhani@LAPTOP-823QCJCF:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 7 7435HS
CPU family: 25
Model: 68
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
```

```
senjaramadhani@LAPTOP-823QCJCF:~$ ps aux | head -20
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.2 165788 10752 ?        Ss   22:32   0:00 /sbin/init
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root     107  0.0  0.3 25672 13360 ?        Ss   22:32   0:00 /lib/systemd/systemd-resolved
systemd+ 126  0.0  0.1 89364  7168 ?        Ssl  22:32   0:00 /lib/systemd/systemd-timesyncd
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message+ 189  0.0  0.1   8580  4608 ?        Ss   22:32   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
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root     197  0.0  0.1 15328  7424 ?        Ss   22:32   0:00 /lib/systemd/systemd-logind
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root    226  0.0  0.0   3196  2176 tty1     Ss+  22:32   0:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
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gnal
root    281  0.0  0.0   3076   896 ?        Ss   22:32   0:00 /init
root    282  0.0  0.0   3092  1024 ?        S   22:32   0:00 /init
senjara+ 283  0.0  0.1 6128  4992 pts/0    Ss   22:32   0:00 -bash
root    284  0.0  0.1  7528  4864 pts/1    Ss   22:32   0:00 /bin/login -f
senjaramadhani@LAPTOP-823QCJCF:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          3.7Gi          362Mi        3.3Gi          3.0Mi        128Mi        3.2Gi
Swap:          1.0Gi           0B          1.0Gi

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CPU op-mode(s): 32-bit, 64-bit
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Vendor ID: AuthenticAMD
Model name: AMD Ryzen 7 7435HS
CPU family: 25
Model: 68
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
```

Praktik 1.2: Program C - Menggunakan Layanan OS

Buat file os_services.c:

```
filter pftreshold v_vmsave_vmload umip vaes vpcmlulqddq rdpid fsmr
Virtualization features:
  Virtualization: AMD-V
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 256 KiB (8 instances)
  L1i: 256 KiB (8 instances)
  L2: 4 MiB (8 instances)
  L3: 16 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-15
Vulnerabilities:
  Gather data sampling: Not affected
  I1b multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Reg file data sampling: Not affected
  Retbleed: Not affected
  Spec rstack overflow: Vulnerable: Safe RET, no microcode
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Retpolines; IBPB conditional; IBRS_FW; STIBP always-on; RSB filling; PBRSE-eIBRS Not affected; BHI Not affected
  Srbds: Not affected
  Tsx async abort: Not affected
senjaramadhani@LAPTOP-823QCJCF:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda 8:0 0 388.4M 1 disk
sdb 8:16 0 186M 1 disk
sdc 8:32 0 1G 0 disk [SWAP]
sdd 8:48 0 1T 0 disk
sde 8:64 0 1T 0 disk /mnt/WSLg/distro
senjaramadhani@LAPTOP-823QCJCF:~$ nano os_services.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o os_services os_services.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
senjaramadhani@LAPTOP-823QCJCF:~$ sudo apt install gcc
```

```
Reading package lists... Done
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o os_services os_services.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
senjaramadhani@LAPTOP-823QCJCF:~$ sudo apt install gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cpp gcc-11 gcc-11-base gcc-12-base libasan6 libatomic1 libc-dev-bin libc-devtools libc6 libc6-dev libc6-i386 libcrypt-dev libgcc-11-dev libgcc-s1
  libgdb3 libgomp1 libisl23 libitm1 liblsan0 libmpc3 libnsl-dev libquadmath0 libstdc++6 libtirpc-dev libtsan0 libubsan1 libxpm4 linux-libc-dev manpages-dev
  rpcsvc-proto
Suggested packages:
  cpp-doc gcc-11-locales gcc-multilib make autoconf automake libtool flex bison gdb gcc-doc gcc-11-multilib gcc-11-doc glibc-doc libgd-tools
Recommended packages:
  libnss-nis libnss-nisplus
The following NEW packages will be installed:
  cpp gcc-11 gcc-11-base gcc-12-base libasan6 libatomic1 libc-dev-bin libc-devtools libc6-dev libc6-i386 libcrypt-dev libgcc-11-dev libgdb3 libgomp1 libisl23
  libitm1 liblsan0 libmpc3 libnsl-dev libquadmath0 libtirpc-dev libtsan0 libubsan1 libxpm4 linux-libc-dev manpages-dev rpcsvc-proto
The following packages will be upgraded:
  gcc-12-base libc6 libgcc-s1 libstdc++6
4 upgraded, 28 newly installed, 0 to remove and 121 not upgraded.
Need to get 47.1 MB/50.9 MB of archives.
After this operation, 147 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6 amd64 2.35-0ubuntu3.11 [3235 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-12-base amd64 12.3.0-1ubuntu1-22.04.2 [20.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libstdc++6 amd64 12.3.0-1ubuntu1-22.04.2 [699 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-s1 amd64 12.3.0-1ubuntu1-22.04.2 [53.9 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11-base amd64 11.4.0-1ubuntu1-22.04.2 [20.8 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 cpp-11 amd64 11.4.0-1ubuntu1-22.04.2 [10.0 MB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-i386 amd64 12.3.0-1ubuntu1-22.04.2 [48.3 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgomp1 amd64 12.3.0-1ubuntu1-22.04.2 [127 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libitm1 amd64 12.3.0-1ubuntu1-22.04.2 [30.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libatomic1 amd64 12.3.0-1ubuntu1-22.04.2 [10.4 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasan6 amd64 11.4.0-1ubuntu1-22.04.2 [2283 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 liblsan0 amd64 12.3.0-1ubuntu1-22.04.2 [1069 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtsan0 amd64 11.4.0-1ubuntu1-22.04.2 [2262 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libubsan1 amd64 12.3.0-1ubuntu1-22.04.2 [976 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libquadmath0 amd64 12.3.0-1ubuntu1-22.04.2 [154 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-11-dev amd64 11.4.0-1ubuntu1-22.04.2 [2517 kB]
```

```

Preparing to unpack .../26-libc6-dev_2.35-0ubuntu3.11_amd64.deb ...
Unpacking libc6-dev:amd64 (2.35-0ubuntu3.11) ...
Selecting previously unselected package manpages-dev.
Preparing to unpack .../27-manpages-dev_5.10-1ubuntu1_all.deb ...
Unpacking manpages-dev (5.10-1ubuntu1) ...
Setting up gcc-11-base:amd64 (11.4.0-1ubuntu1-22.04.2) ...
Setting up manpages-dev (5.10-1ubuntu1) ...
Setting up libxpm4:amd64 (1:3.5.12-1ubuntu0.22.04.2) ...
Setting up linux-libc-dev:amd64 (5.15.0-157.167) ...
Setting up libgomp1:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libasan6:amd64 (11.4.0-1ubuntu1-22.04.2) ...
Setting up libtirpc-dev:amd64 (1.3.2-2ubuntu0.1) ...
Setting up rpcsvc-proto (1.4.2-0ubuntu6) ...
Setting up libquadmath0:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libgd3:amd64 (2.3.0-2ubuntu2.3) ...
Setting up libmpc3:amd64 (1.2.1-2build1) ...
Setting up libatomic1:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libubsan1:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libisl-dev:amd64 (1.3.0-2build2) ...
Setting up libcrypt-dev:amd64 (1:4.4.27-1) ...
Setting up libisl23:amd64 (0.24-2build1) ...
Setting up libc-dev-bin (2.35-0ubuntu3.11) ...
Setting up libcc1-0:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up liblsan0:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libitm1:amd64 (12.3.0-1ubuntu1-22.04.2) ...
Setting up libc-devtools (2.35-0ubuntu3.11) ...
Setting up libtsan0:amd64 (11.4.0-1ubuntu1-22.04.2) ...
Setting up cpp-11 (11.4.0-1ubuntu1-22.04.2) ...
Setting up libgcc-11-dev:amd64 (11.4.0-1ubuntu1-22.04.2) ...
Setting up gcc-11 (11.4.0-1ubuntu1-22.04.2) ...
Setting up cpp (4:11.2.0-1ubuntu1) ...
Setting up libc6-dev:amd64 (2.35-0ubuntu3.11) ...
Setting up gcc (4:11.2.0-1ubuntu1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
senjaramadhani@LAPTOP-823QCJCF:~$ ./os_services
-bash: ./os_services: No such file or directory
senjaramadhani@LAPTOP-823QCJCF:~$ ls
find ~ -name os_services
os_services.c

```

2. User and Operating-System Interface

Praktik 2.1: Command Line Interface (CLI)

```

senjaramadhani@LAPTOP-823QCJCF:~$ pwd
/home/senjaramadhani
senjaramadhani@LAPTOP-823QCJCF:~$ ls -la
total 36
drwxr-xr-x 4 senjaramadhani senjaramadhani 4096 Oct 14 23:15 .
drwxr-xr-x 3 root root 4096 Oct 14 22:32 ..
-rw-r--r-- 1 senjaramadhani senjaramadhani 5493 Oct 14 23:15 .bash_history
-rw-r--r-- 1 senjaramadhani senjaramadhani 220 Oct 14 22:32 .bash_logout
-rw-r--r-- 1 senjaramadhani senjaramadhani 3771 Oct 14 22:32 .bashrc
drwx----- 2 senjaramadhani senjaramadhani 4096 Oct 14 22:32 .cache
drwxr-xr-x 3 senjaramadhani senjaramadhani 4096 Oct 14 22:32 .local
-rw-r--r-- 1 senjaramadhani senjaramadhani 0 Oct 14 22:32 .motd_shown
-rw-r--r-- 1 senjaramadhani senjaramadhani 807 Oct 14 22:32 .profile
-rw-r--r-- 1 senjaramadhani senjaramadhani 0 Oct 14 22:51 .sudo_as_admin_successful
-rw-r--r-- 1 senjaramadhani senjaramadhani 1321 Oct 14 22:50 os_services.c
senjaramadhani@LAPTOP-823QCJCF:~$ mkdir test_dir
senjaramadhani@LAPTOP-823QCJCF:~$ cd test_dir
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ ls -l | grep "ad"
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ ps aux | grep bash | wc -l
3
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ echo "Hello World" > output.txt
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ cat output.txt
Hello World
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ cat < output.txt
Hello World
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ sleep 30 &
[1] 1895
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ jobs
[1]+  Running                  sleep 30 &
senjaramadhani@LAPTOP-823QCJCF:~/test_dir$ fg %1
sleep 30

```

Praktik 2.2: Program C - Shell Sederhana

Buat file simple_shell.c:

```

senjaramadhani@LAPTOP-823QCJCF:~$ nano simple_shell.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o simple_shell simple_shell.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./simple_shell
myshell> ^C
senjaramadhani@LAPTOP-823QCJCF:~$ |

```

3. System Calls

Praktik 3.1: Process Control System Calls

Buat file process_syscalls.c:

Praktik 3.2: File Management System Calls

Buat file file_syscalls.c:

Praktik 3.3: Information Maintenance System Calls

Buat file info_syscalls.c:

```
senjaramadhani@LAPTOP-823QCJCF:~$ nano process_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ nano file_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ nano info_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o process_syscalls process_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o file_syscalls file_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o info_syscalls info_syscalls.c
senjaramadhani@LAPTOP-823QCJCF:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 1442
Parent waiting for child (PID: 1443)...
Child PID: 1443
Child's Parent PID: 1442
Child executing 'ls' command...
total 92
-rwxr-xr-x 1 senjaramadhani senjaramadhani 16352 Oct 14 23:27 file_syscalls
-rw-r--r-- 1 senjaramadhani senjaramadhani 819 Oct 14 23:25 file_syscalls.c
-rwxr-xr-x 1 senjaramadhani senjaramadhani 16384 Oct 14 23:27 info_syscalls
-rw-r--r-- 1 senjaramadhani senjaramadhani 625 Oct 14 23:27 info_syscalls.c
-rw-r--r-- 1 senjaramadhani senjaramadhani 1321 Oct 14 22:50 os_services.c
-rwxr-xr-x 1 senjaramadhani senjaramadhani 16360 Oct 14 23:27 process_syscalls
-rw-r--r-- 1 senjaramadhani senjaramadhani 720 Oct 14 23:24 process_syscalls.c
-rwxr-xr-x 1 senjaramadhani senjaramadhani 16528 Oct 14 23:21 simple_shell
-rw-r--r-- 1 senjaramadhani senjaramadhani 793 Oct 14 23:21 simple_shell.c
drwxr-xr-x 2 senjaramadhani senjaramadhani 4096 Oct 14 23:17 test_dir
Child exited with status: 0
senjaramadhani@LAPTOP-823QCJCF:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
senjaramadhani@LAPTOP-823QCJCF:~$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 1445
Parent Process ID: 1403
User ID: 1000
```

```
senjaramadhani@LAPTOP-823QCJCF:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
senjaramadhani@LAPTOP-823QCJCF:~$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 1445
Parent Process ID: 1403
User ID: 1000
Group ID: 1000
Current time: Tue Oct 14 23:27:52 2025
Clock ticks since boot: 1718362610
Sleeping for 2 seconds...
Awake!
senjaramadhani@LAPTOP-823QCJCF:~$ |
```

4. System Services

Praktik 4.1: Menggunakan system

```
senjaramadhani@LAPTOP-823QCJCF:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
apport.service                     loaded active exited  LSB: automatic crash report generation
console-getty.service              loaded active running  Console Getty
console-setup.service              loaded active exited  Set console font and keymap
cron.service                       loaded active running  Regular background program processing daemon
dbus.service                      loaded active running  D-Bus System Message Bus
getty@tty1.service                 loaded active running  Getty on tty1
keyboard-setup.service             loaded active exited  Set the console keyboard layout
kmod-static-nodes.service          loaded active exited  Create List of Static Device Nodes
networkd-dispatcher.service        loaded active running  Dispatcher daemon for systemd-networkd
packagekit.service               loaded active running  PackageKit Daemon
plymouth-quit-wait.service         loaded active exited  Hold until boot process finishes up
plymouth-quit.service             loaded active exited  Terminate Plymouth Boot Screen
plymouth-read-write.service        loaded active exited  Tell Plymouth To Write Out Runtime Data
polkit.service                    loaded active running  Authorization Manager
rsyslog.service                   loaded active running  System Logging Service
setvtrgb.service                  loaded active exited  Set console scheme
snapd.seeded.service              loaded active exited  Wait until snapd is fully seeded
ssh.service                       loaded active running  OpenBSD Secure Shell server
systemd-journal-flush.service      loaded active exited  Flush Journal to Persistent Storage
systemd-journald.service           loaded active running  Journal Service
systemd-logind.service             loaded active running  User Login Management
systemd-remount-fs.service         loaded active exited  Remount Root and Kernel File Systems
systemd-resolved.service           loaded active running  Network Name Resolution
systemd-sysctl.service             loaded active exited  Apply Kernel Variables
systemd-sysusers.service           loaded active exited  Create System Users
systemd-timesyncd.service          loaded active running  Network Time Synchronization
systemd-tmpfiles-setup-dev.service loaded active exited  Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service     loaded active exited  Create Volatile Files and Directories
systemd-udev-trigger.service       loaded active exited  Coldplug All udev Devices
systemd-udevd.service              loaded active running  Rule-based Manager for Device Events and Files
systemd-update-utmp.service        loaded active exited  Record System Boot/Shutdown in UTMP
systemd-user-sessions.service      loaded active exited  Permit User Sessions
ufw.service                       loaded active exited  Uncomplicated firewall
unattended-upgrades.service        loaded active running  Unattended Upgrades Shutdown
user-runtime-dir@1000.service      loaded active exited  User Runtime Directory /run/user/1000
user@1000.service                 loaded active running  User Manager for UID 1000

LOAD = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
```

```

senjaramadhani@LAPTOP-823QCJCF:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-10-14 23:37:59 WIB; 1min 26s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 2149 (sshd)
    Tasks: 1 (limit: 4574)
   Memory: 1.7M
      CPU: 25ms
   CGroup: /system.slice/ssh.service
           └─2149 /usr/sbin/sshd -D [listener] 0 of 10-100 startups

Oct 14 23:37:59 LAPTOP-823QCJCF systemd[1]: Starting OpenBSD Secure Shell server...
Oct 14 23:37:59 LAPTOP-823QCJCF sshd[2149]: Server listening on 0.0.0.0 port 22.
Oct 14 23:37:59 LAPTOP-823QCJCF sshd[2149]: Server listening on :: port 22.
Oct 14 23:37:59 LAPTOP-823QCJCF systemd[1]: Started OpenBSD Secure Shell server.
senjaramadhani@LAPTOP-823QCJCF:~$ journalctl -u ssh -n 20
Oct 14 23:37:59 LAPTOP-823QCJCF systemd[1]: Starting OpenBSD Secure Shell server...
Oct 14 23:37:59 LAPTOP-823QCJCF sshd[2149]: Server listening on 0.0.0.0 port 22.
Oct 14 23:37:59 LAPTOP-823QCJCF sshd[2149]: Server listening on :: port 22.
Oct 14 23:37:59 LAPTOP-823QCJCF systemd[1]: Started OpenBSD Secure Shell server.
senjaramadhani@LAPTOP-823QCJCF:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    VENDOR PRESET
apparmor.service                        enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                    enabled enabled
cloud-init-local.service               enabled enabled
cloud-init.service                     enabled enabled
console-setup.service                  enabled enabled
cron.service                           enabled enabled
dmesg.service                          enabled enabled
e2scrub_reap.service                   enabled enabled
getty@.service                         enabled enabled
irqbalance.service                    enabled enabled
keyboard-setup.service                 enabled enabled
landscape-client.service               enabled enabled
networkd-dispatcher.service            enabled enabled
rsyslog.service                        enabled enabled
setvtrgb.service                       enabled enabled

```

```

senjaramadhani@LAPTOP-823QCJCF:~$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    VENDOR PRESET
apparmor.service                        enabled enabled
cloud-config.service                   enabled enabled
cloud-final.service                    enabled enabled
cloud-init-local.service               enabled enabled
cloud-init.service                     enabled enabled
console-setup.service                  enabled enabled
cron.service                           enabled enabled
dmesg.service                          enabled enabled
e2scrub_reap.service                   enabled enabled
getty@.service                         enabled enabled
irqbalance.service                    enabled enabled
keyboard-setup.service                 enabled enabled
landscape-client.service               enabled enabled
networkd-dispatcher.service            enabled enabled
rsyslog.service                        enabled enabled
setvtrgb.service                       enabled enabled
snapd.apparmor.service                 enabled enabled
snapd.autoimport.service               enabled enabled
snapd.core-fixup.service                enabled enabled
snapd.recovery-chooser-trigger.service enabled enabled
snapd.seeded.service                  enabled enabled
snapd.service                          enabled enabled
snapd.system-shutdown.service          enabled enabled
ssh.service                            enabled enabled
systemd-pstore.service                 enabled enabled
systemd-resolved.service               enabled enabled
systemd-timesyncd.service              enabled enabled
ua-reboot-cmds.service                 enabled enabled
ubuntu-advantage.service               enabled enabled
ufw.service                            enabled enabled
unattended-upgrades.service            enabled enabled

31 unit files listed.
senjaramadhani@LAPTOP-823QCJCF:~$

```

Praktik 4.2: Membuat Custom System Service

Buat program sederhana my_daemon.c:

```

senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ nano my_daemon.c
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ gcc -o my_daemon my_daemon.c
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ chmod +x my_daemon
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ ls -l ~/my_daemon_dir/my_daemon
-rwxr-xr-x 1 senjaramadhani senjaramadhani 16096 Oct 14 23:58 /home/senjaramadhani/my_daemon_dir/my_daemon
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo nano /etc/systemd/system/my_daemon.service
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo systemctl daemon-reload
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo systemctl start my_daemon
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo systemctl status my_daemon
● my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-10-14 23:59:33 WIB; 7s ago
     Main PID: 2467 (my_daemon)
    Tasks: 1 (limit: 4574)
   Memory: 180.0K
      CPU: 1ms
   CGroup: /system.slice/my_daemon.service
           └─2467 /home/senjaramadhani/my_daemon_dir/my_daemon

Oct 14 23:59:33 LAPTOP-823QCJCF systemd[1]: Started My Custom Daemon.
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo journalctl -u my_daemon
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[1]: Starting My Custom Daemon...
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[2262]: my_daemon.service: Failed to locate executable /path/to/my_daemon: No such file or directory
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[2262]: my_daemon.service: Failed at step EXEC spawning /path/to/my_daemon: No such file or directory
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 14 23:45:57 LAPTOP-823QCJCF systemd[1]: Failed to start My Custom Daemon.
Oct 14 23:45:58 LAPTOP-823QCJCF systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.

senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo systemctl stop my_daemon
senjaramadhani@LAPTOP-823QCJCF:~/my_daemon_dir$ sudo nano /etc/systemd/system/my_daemon.service

```

5. Linkers and Loaders

Praktik 5.1: Proses Linking

Buat tiga file:

math_ops.h:

math_ops.c:

main.c:

```
senjaramadhani@LAPTOP-823QCJCF:~$ nano math_ops.h
senjaramadhani@LAPTOP-823QCJCF:~$ nano math_ops.c
senjaramadhani@LAPTOP-823QCJCF:~$ nano main.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -E main.c -o main.i
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -E math_ops.c -o math_ops.i
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -S main.c -o main.s
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -S math_ops.c -o math_ops.s
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -e main.c -o main.o
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -c math_ops.c -o math_ops.o
senjaramadhani@LAPTOP-823QCJCF:~$ gcc main.o math_ops.o -o program
senjaramadhani@LAPTOP-823QCJCF:~$ nm main.o
U add
0000000000000000 T main
U multiply
U printf
senjaramadhani@LAPTOP-823QCJCF:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
senjaramadhani@LAPTOP-823QCJCF:~$ nm program
0000000000003dc0 d _DYNAMIC
0000000000003fb0 d _GLOBAL_OFFSET_TABLE_
0000000000002000 R _IO_stdin_used
w _ITM_deregisterTMCloneTable
w _ITM_registerTMCloneTable
0000000000002148 r __FRAME_END__
000000000000201c r __GNU_EH_FRAME_HDR
0000000000004010 D __TMC_END__
000000000000038c r __abi_tag
0000000000004010 B __bss_start
w __cxa_finalize@GLIBC_2.2.5
0000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
0000000000003dc0 d __do_global_dtors_aux_fini_array_entry
0000000000004000 D __dso_handle
0000000000003db8 d __frame_dummy_init_array_entry
w __gmon_start__
U __libc_start_main@GLIBC_2.34
0000000000003db8 d __frame_dummy_init_array_entry
w __gmon_start__
U __libc_start_main@GLIBC_2.34
0000000000004010 D _edata
0000000000004010 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
0000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
senjaramadhani@LAPTOP-823QCJCF:~$ ldd program
linux-vdso.so.1 (0x00007ffffd083df000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000072010e600000)
/lib64/ld-linux-x86-64.so.2 (0x000072010e94e000)
```

Praktik 5.2: Static vs Dynamic Linking

```
senjaramadhani@LAPTOP-823QCJCF:~$ gcc main.c math_ops.c -o program_static -static
senjaramadhani@LAPTOP-823QCJCF:~$ gcc main.c math_ops.c -o program_dynamic
senjaramadhani@LAPTOP-823QCJCF:~$ ldd program_static
not a dynamic executable
senjaramadhani@LAPTOP-823QCJCF:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffffd083df000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000072010e600000)
/lib64/ld-linux-x86-64.so.2 (0x000072010e94e000)
senjaramadhani@LAPTOP-823QCJCF:~$ |
```

Praktik 5.3: Membuat dan Menggunakan Shared Library

```
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -c -fPIC math_ops.c -o math_ops.o
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -shared -o libmathops.so math_ops.o
senjaramadhani@LAPTOP-823QCJCF:~$ gcc main.c -L. -lmathops -o program_shared
senjaramadhani@LAPTOP-823QCJCF:~$ LD_LIBRARY_PATH=. ./program_shared
5 * 3 = 0
5 * 3 = 15
senjaramadhani@LAPTOP-823QCJCF:~$ readelf -d program_shared

Dynamic section at offset 0x2da0 contains 28 entries:
Tag              Type              Name/Value
0x0000000000000001 (NEEDED)          Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)          Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x1000
0x000000000000000d (FINI)            0x1104
0x0000000000000019 (INIT_ARRAY)       0x3d98
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)        0x3da0
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x0000000000000005 (GNU_HASH)          0x3db0
0x0000000000000006 (SYMtab)             0x3db0
0x000000000000000a (STRtab)             0x3db0
0x0000000000000006 (SYMtab)             0x3db0
0x000000000000000a (STRtab)             0x3db0
0x000000000000000b (SYMtab)             0x3db0
0x0000000000000015 (DEBUG)              0x0
0x0000000000000003 (PLTgot)             0x3fab
0x0000000000000002 (PLTrelsz)            72 (bytes)
0x0000000000000014 (PLTrel)              REL
0x0000000000000017 (PLTrel)              REL
0x0000000000000007 (RELA)                0x5a0
0x0000000000000008 (RELA)                192 (bytes)
0x0000000000000009 (RELA)                24 (bytes)
0x000000000000001e (FLAGS)               BIND_NOW
0x000000000000001f (FLAGS_1)            Flags: NOW PIE
0x0000000000000000 (VERNEED)            0x370
0x0000000000000000 (VERNEEDNUM)         1
0x0000000000000000 (VERSYM)             0x55a
0x0000000000000000 (RELACOUNT)          3
0x0000000000000000 (NULL)              0x0
senjaramadhani@LAPTOP-823QCJCF:~$ |
```

6. Why Applications Are Operating-System Specific

Praktik 6.1: System Call Differences

Buat file os_specific.c:

Praktik 6.2: Binary Format Analysis

```
senjaramadhani@LAPTOP-823QCJCF:~$ nano os_specific.c
senjaramadhani@LAPTOP-823QCJCF:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=84e21dd488c5917
3b93ecce3c9566e83f498d5f3, for GNU/Linux 3.2.0, not stripped
senjaramadhani@LAPTOP-823QCJCF:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:       ELF64
  Data:        2's complement, little endian
  Version:     1 (current)
  OS/ABI:      UNIX - System V
  ABI Version: 0
  Type:        DYN (Position-Independent Executable file)
  Machine:     Advanced Micro Devices X86-64
  Version:     0x1060
  Entry point address: 0x1060
  Start of program headers: 64 (bytes into file)
  Start of section headers: 14072 (bytes into file)
  Flags:       0x0
  Size of this header: 64 (bytes)
  Size of program headers: 56 (bytes)
  Number of program headers: 13
  Size of section headers: 64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
senjaramadhani@LAPTOP-823QCJCF:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:

Section Headers:
 [Nr] Name              Type              Address            Offset
     Size              EntSize          Flags              Link      Info      Align
 [ 0]                     NULL              0000000000000000  00000000
     0000000000000000  0000000000000000  0 0 0
 [ 1] .interp             PROGBITS          0000000000000318  00000318
     000000000000001c  0000000000000000  A 0 0 1
 [ 2] .note.gnu.pr[...]   NOTE              0000000000000338  00000338
     0000000000000030  0000000000000000  A 0 0 8
 [ 3] .note.gnu.bu[...]   NOTE              0000000000000368  00000368
     0000000000000024  0000000000000000  A 0 0 4
 [ 4] .note.ABI-tag       NOTE              000000000000038c  0000038c
     000000000000011a  0000000000000000  0 0 1

Key to Flags:
  W (write), A (alloc), X (execute), M (merge), S (strings), I (info),
  L (link order), O (extra OS processing required), G (group), T (TLS),
  C (compressed), x (unknown), o (OS specific), E (exclude),
  D (mbind), l (large), p (processor specific)
senjaramadhani@LAPTOP-823QCJCF:~$ readelf -l program
Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:
  Type           Offset             VirtAddr           PhysAddr
                 FileSiz            MemSiz              Flags             Align
PHDR             0x0000000000000040 0x0000000000000040 0x0000000000000040
INTERP           0x00000000000002d8 0x00000000000002d8 R                 0x8
LOAD             0x0000000000000318 0x0000000000000318 0x0000000000000318
LOAD             0x000000000000001c 0x000000000000001c 0x000000000000001c R                 0x1
                 [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD             0x0000000000000000 0x0000000000000000 0x0000000000000000
LOAD             0x0000000000000628 0x0000000000000628 R                 0x1000
LOAD             0x0000000000000100 0x0000000000000100 0x0000000000000100
LOAD             0x00000000000001e1 0x00000000000001e1 R E               0x1000
LOAD             0x0000000000000200 0x0000000000000200 0x0000000000000200
LOAD             0x000000000000014c 0x000000000000014c R                 0x1000
LOAD             0x00000000000002db8 0x00000000000002db8 0x00000000000002db8
LOAD             0x0000000000000258 0x0000000000000258 0x0000000000000258 RW              0x1000
DYNAMIC          0x00000000000002dc8 0x00000000000002dc8 0x00000000000002dc8
LOAD             0x00000000000001f0 0x00000000000001f0 RW                 0x8
NOTE            0x0000000000000338 0x0000000000000338 0x0000000000000338
NOTE            0x0000000000000030 0x0000000000000030 R                 0x8
NOTE            0x0000000000000368 0x0000000000000368 0x0000000000000368
GNU_PROPERTY     0x0000000000000044 0x0000000000000044 R                 0x4
GNU_PROPERTY     0x0000000000000338 0x0000000000000338 0x0000000000000338
GNU_PROPERTY     0x0000000000000030 0x0000000000000030 R                 0x8
GNU_EH_FRAME    0x0000000000000201c 0x0000000000000201c 0x0000000000000201c
10 .eh_frame_hdr
11
12 .init_array .fini_array .dynamic .got
senjaramadhani@LAPTOP-823QCJCF:~$ objdump -d program | head -50
program:          file format elf64-x86-64

Disassembly of section .init:
0000000000001000 <.init>:
1000: f3 0f 1e fa          endbr64
1004: 48 83 ec 08          sub    $0x8,%rsp
1008: 48 8b 05 d9 2f 00 00 mov    0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
100f: 48 85 c0             test   %rax,%rax
1012: 74 02              je     1016 <.init+0x16>
1014: ff d0              call   *%rax
1016: 48 83 c4 08          add    $0x8,%rsp
101a: c3                 ret

Disassembly of section .plt:
0000000000001020 <.plt>:
1020: ff 35 9a 2f 00 00    push  0x2f9a(%rip)             # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>
1026: f2 ff 25 9b 2f 00 00 bnd jmp *0x2f9b(%rip)          # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
102d: 0f 1f 00             nopl   (%rax)
1030: f3 0f 1e fa          endbr64
1034: 68 00 00 00 00 00    push  $0x0
1039: f2 e9 e1 ff ff ff    bnd jmp 1020 <.init+0x20>
103f: 90                 nop

Disassembly of section .plt.got:
0000000000001040 <__cxa_finalize@plt>:
1040: 0f 1f 00             nopl   (%rax)
```

7. OS Design and Implementation

Praktik 7.1: Kernel Module Sederhana

Buat file hello_module.c:

```
senjaramadhani@LAPTOP-823QCJCF:~$ nano hello_module.c
senjaramadhani@LAPTOP-823QCJCF:~$ nano Makefile
senjaramadhani@LAPTOP-823QCJCF:~$ make
make: *** No targets. Stop.
senjaramadhani@LAPTOP-823QCJCF:~$ sudo insmod hello_module.ko
insmod: ERROR: could not insert module hello_module.ko: Invalid module format
senjaramadhani@LAPTOP-823QCJCF:~$ lsmod | grep hello
senjaramadhani@LAPTOP-823QCJCF:~$ dmesg | tail
[ 2.733035] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 2.733060] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 2.901685] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 4.272501] systemd-journald[61]: File /var/log/journal/4d24d7d873da4319ae2ce7ab31678dea/user-1000.journal corrupted or uncleanly shut down, renaming and replacing.
[ 6.136483] WSL (260) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 12.581154] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 48.447600] hv_balloon: Max. dynamic memory size: 4004 MB
[ 621.081125] mini_init (244): drop_caches: 1
[ 3056.061574] Invalid ELF header magic: != \x7fELF
[ 3199.429301] Invalid ELF header magic: != \x7fELF
senjaramadhani@LAPTOP-823QCJCF:~$
senjaramadhani@LAPTOP-823QCJCF:~$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
senjaramadhani@LAPTOP-823QCJCF:~$ dmesg | tail
[ 2.733035] kvm_amd: kvm_amd: Hyper-V Direct TLB Flush enabled
[ 2.733060] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 2.901685] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 4.272501] systemd-journald[61]: File /var/log/journal/4d24d7d873da4319ae2ce7ab31678dea/user-1000.journal corrupted or uncleanly shut down, renaming and replacing.
[ 6.136483] WSL (260) ERROR: CheckConnection: getaddrinfo() failed: -5
[ 12.581154] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 48.447600] hv_balloon: Max. dynamic memory size: 4004 MB
[ 621.081125] mini_init (244): drop_caches: 1
[ 3056.061574] Invalid ELF header magic: != \x7fELF
[ 3199.429301] Invalid ELF header magic: != \x7fELF
senjaramadhani@LAPTOP-823QCJCF:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
senjaramadhani@LAPTOP-823QCJCF:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
senjaramadhani@LAPTOP-823QCJCF:~$ sysctl -a | head -20
sysctl: abi.vsyscall32 = 1
permission denied on key 'kernel.cad_pid' debug.exception-trace = 1
senjaramadhani@LAPTOP-823QCJCF:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
senjaramadhani@LAPTOP-823QCJCF:~$ sysctl -a | head -20
sysctl: abi.vsyscall32 = 1
permission denied on key 'kernel.cad_pid' debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autosject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCN:
dev.cdrom.info = Reports media changed:
senjaramadhani@LAPTOP-823QCJCF:~$ dmesg | tail -30
[ 2.318290] systemd-journald[61]: Received client request to flush runtime journal.
[ 2.327942] systemd-journald[61]: File /var/log/journal/4d24d7d873da4319ae2ce7ab31678dea/system.journal corrupted or uncleanly shut down, renaming and replacing.
[ 2.403258] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.465001] ACPI: AC Adapter [AC] (on-line)
[ 2.509469] ACPI: battery: Slot [BAT1] (battery present)
[ 2.711192] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.711766] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.712245] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.712679] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.713112] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.713590] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.714011] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.714442] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.715069] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.715971] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.716388] misc dkg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.732900] kvm_amd: TSC scaling supported
[ 2.732994] kvm_amd: Nested Virtualization enabled
[ 2.732997] kvm_amd: Nested Paging enabled
senjaramadhani@LAPTOP-823QCJCF:~$ lsmod
Module                  Size  Used by
tls                     98304  0
intel_rapl_msr          16384  0
intel_rapl_common       32768  1 intel_rapl_msr
kvm_amd                 126976  0
kvm                     970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel            16384  0
battery                 20480  0
ac                      16384  0
sch_fq_codel            16384  1
configs                 53248  0
autofs4                 45056  0
br_netfilter            28672  0
bridge                  282624  1 br_netfilter
stp                     12288  1 bridge
llc                     12288  2 bridge,stp
ip_tables               28672  0
tun                     53248  0
senjaramadhani@LAPTOP-823QCJCF:~$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3
senjaramadhani@LAPTOP-823QCJCF:~$ |
```

Praktik 8.1: Eksplorasi Struktur Layering

[illegible]

```

senjaramadhani@LAPTOP-823QCJCF:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.9  0.2 165936 10624 ?        Ss   06:04   0:00 /sbin/init
root         2  0.0  0.0   3072  1792 ?        Sl   06:04   0:00 /init
root         7  0.0  0.0   3008  1920 ?        Sl   06:04   0:00 plan9 --control-socket /7 --log-level 4 --server-fd 8
root        64  0.2  0.2  31412 10880 ?        Ss<  06:04   0:00 /lib/systemd/systemd-journald
root        89  0.2  0.1  22840  5888 ?        Ss   06:04   0:00 /lib/systemd/systemd-udev
systemd+   104  0.1  0.3  25672 13292 ?        Ss   06:04   0:00 /lib/systemd/systemd-resolved
systemd+   111  0.1  0.1  89364  7040 ?        Ssl  06:04   0:00 /lib/systemd/systemd-timesyncd
root       189  0.0  0.0   4308  2688 ?        Ss   06:04   0:00 /usr/sbin/cron -f -p
message+   191  0.0  0.1   8584  4352 ?        Ss   06:04   0:00 @dbus-daemon --system --address=systemd: --nofork --n
root       196  0.0  0.0   2776  1280 ?        Ss   06:04   0:00 /home/senjaramadhani/my_daemon_dir/my_daemon
root       197  0.2  0.4  30136 18816 ?        Ss   06:04   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s
syslog     198  0.0  0.1  222404  4992 ?        Ssl  06:04   0:00 /usr/sbin/rsyslogd -n -iNONE
root       202  0.1  0.1  15328  7424 ?        Ss   06:04   0:00 /lib/systemd/systemd-logind
root       224  0.0  0.0   3240  2176 hvc0     Ss+  06:04   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud consol
root       226  0.0  0.2  15436  8960 ?        Ss   06:04   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startu
root       227  0.0  0.0   3196  2176 tty1     Ss+  06:04   0:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root       229  0.1  0.5 107164 21248 ?        Ssl  06:04   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt
root       296  0.0  0.0   3076   896 ?        Ss   06:04   0:00 /init
root       297  0.0  0.0   3092  1024 ?        R    06:04   0:00 /init
senjara+   298  0.0  0.1   6124  4992 pts/0    Ss   06:04   0:00 -bash
root       299  0.0  0.1   7520  4736 pts/1    Ss   06:04   0:00 /bin/login -f
senjara+   343  0.0  0.1   6104  4992 pts/1    S+   06:04   0:00 -bash
senjara+   383  0.0  0.0   7484  3072 pts/0    R+   06:05   0:00 ps aux
senjaramadhani@LAPTOP-823QCJCF:~$ top
top - 06:05:55 up 4:30, 1 user, load average: 0.07, 0.09, 0.04
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3822.2 total, 3322.5 free, 372.4 used, 127.3 buff/cache
MiB Swap: 1024.0 total, 1024.0 free,  0.0 used, 3318.4 avail Mem

   PID USER      PR  NI   VIRT    RES    SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0 165936 10624  8064  S   0.0   0.3   0:00.74 systemd
    2 root        20   0  3072    1792   1792  S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0  3088    1920   1920  S   0.0   0.0   0:00.00 init
   64 root       19  -1  31412 10880  10112  S   0.0   0.3   0:00.19 systemd-journal
   89 root       20   0  22840  5888   4608  S   0.0   0.2   0:00.17 systemd-udev
  104 systemd+   20   0  25672 13292  8960  S   0.0   0.3   0:00.12 systemd-resolve

senjaramadhani@LAPTOP-823QCJCF:~$ which bash
/usr/bin/bash
senjaramadhani@LAPTOP-823QCJCF:~$ which gcc
/usr/bin/gcc
senjaramadhani@LAPTOP-823QCJCF:~$ |

```

Praktik 8.3: Program - Virtual File System

Buat file procfs_read.c:

```

senjaramadhani@LAPTOP-823QCJCF:~$ which gcc
/usr/bin/gcc
senjaramadhani@LAPTOP-823QCJCF:~$ nano procfs_read.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:32:60: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   32 |     snprintf(proc_path, sizeof(proc_path), "/proc/%d/status", getpid());
      |                                         ^
senjaramadhani@LAPTOP-823QCJCF:~$ ./procfs_read
=== CPU Information ===
processor       : 0
vendor_id     : AuthenticAMD
cpu family    : 25
model         : 68
model name    : AMD Ryzen 7 7435HS
stepping      : 1
microcode     : 0xffffffff
cpu MHz       : 3094.109
cache size    : 512 KB
physical id   : 0

=== Memory Information ===
MemTotal:     3913916 kB
MemFree:      3353332 kB
MemAvailable: 3363420 kB
Buffers:      2788 kB
Cached:       123936 kB

=== Process Information ===
Name:          procfs_read
Umask:         0022
State:         R (running)
Tgid:          393
Ngid:          0
Pid:           393
PPid:          298
TracerPid:     0
Uid:           1000 1000 1000
Gid:           1000 1000 1000
senjaramadhani@LAPTOP-823QCJCF:~$ |

```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

```
senjaramadhan1@LAPTOP-823QCJCF:~$ dmesg | grep hello_module
senjaramadhan1@LAPTOP-823QCJCF:~$ systemd-analyze
Startup finished in 1.712s (userspace)
graphical.target reached after 1.679s in userspace
senjaramadhan1@LAPTOP-823QCJCF:~$ systemd-analyze blame
746ms landscape-client.service
425ms snapd.seeded.service
397ms dev-sde.device
290ms snapd.service
238ms networkd-dispatcher.service
179ms systemd-udev-trigger.service
163ms systemd-resolved.service
106ms systemd-timesyncd.service
98ms systemd-journald.service
89ms apport.service
85ms systemd-logind.service
77ms ssh.service
65ms keyboard-setup.service
63ms systemd-journal-flush.service
53ms systemd-udev.service
53ms dev-hugepages.mount
52ms dev-mqueue.mount
50ms e2scrub_reap.service
49ms sys-kernel-debug.mount
48ms systemd-tmpfiles-setup.service
46ms sys-kernel-tracing.mount
42ms rsyslog.service
39ms snapd.socket
37ms kmod-static-nodes.service
34ms modprobe@drm.service
31ms modprobe@efi_pstore.service
31ms modprobe@fuse.service
29ms systemd-remount-fs.service
26ms systemd-sysctl.service
lines 1-29
senjaramadhan1@LAPTOP-823QCJCF:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @1.679s
└─multi-user.target @1.678s
   └─snapd.seeded.service @947ms +425ms
      └─basic.target @909ms
         └─sockets.target @908ms
            └─snapd.socket @866ms +39ms
               └─sysinit.target @854ms
                  └─systemd-timesyncd.service @748ms +106ms
                     └─systemd-tmpfiles-setup.service @686ms +48ms
                        └─systemd-journal-flush.service @621ms +63ms
                           └─systemd-journald.service @518ms +98ms
                              └─systemd-journald.socket @466ms
                                 └─system.slice @454ms
                                    └─.slice @454ms
senjaramadhan1@LAPTOP-823QCJCF:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
senjaramadhan1@LAPTOP-823QCJCF:~$ lsinitramfs /boot/initrd.img-${uname -r} | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
senjaramadhan1@LAPTOP-823QCJCF:~$
```

Lihat boot messages

```
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu
Jun 5 18:38:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_E
NABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001023ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[ 0.000000] Hyper-V: Nested features: 0x4a0000
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote I/O flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 3894.109 MHz processor
[ 0.000063] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.000066] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.000069] last_pfn = 0x102400 max_arch_pfn = 0x400000000
[ 0.000093] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.000096] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[ 0.001141] last_pfn = 0xf8000 max_arch_pfn = 0x400000000
[ 0.001160] Using GB pages for direct mapping
[ 0.000273] RAMDISK: [mem 0x00b04000-0x00c7ffff]
[ 0.000277] ACPI: Early table checksum verification disabled
[ 0.000284] ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
[ 0.000289] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.000296] ACPI: FACP 0x0000000000101000 000114 (v06 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.000301] ACPI: DSDT 0x00000000001011B8 01E11C (v02 MSFTVM DSDT01 00000001 INTL 20230628)
[ 0.000303] ACPI: FACS 0x0000000000101114 000040
[ 0.000304] ACPI: OEM0 0x0000000000101154 000064 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
[ 0.000306] ACPI: SRAT 0x000000000011F2D4 0003B0 (v02 VIRTUAL MICROSOFT 00000001 MSFT 00000001)
:]
```

Buat file bootloader_concept.asm (untuk pembelajaran):

```
senjaramadhani@LAPTOP-823QCJCF:~$ cat /proc/cmdline
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
senjaramadhani@LAPTOP-823QCJCF:~$ sudo nano /etc/default/grub
senjaramadhani@LAPTOP-823QCJCF:~$ nano bootloader_concept.asm
senjaramadhani@LAPTOP-823QCJCF:~$
```

Praktik 10.1: Using strace

```

senjaramadhanil@LAPTOP-8230CJCF:~$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffff54cadb20 /* 25 vars */) = 0
brk(NULL)
      = 0x5ef4e4ae18000
arch_prctl(0x3001 /* ARCH.??? */, 0x7ffd9dad69a0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76c2b1ec5000
access("/etc/ld.so.preload", R_OK)
      = -1 ENOENT (No such file or directory)
open(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=21996, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 21996, PROT_READ, MAP_PRIVATE, 3, 0) = 0x76c2b1ebf000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=166280, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 177672, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76c2b1e93000
mprotect(0x76c2b1e93000, 139264, PROT_NONE) = 0
mmap(0x76c2b1e99000, 106496, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x76c2b1e99000
mmap(0x76c2b1ebb000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76c2b1ebb000
mmap(0x76c2b1ebb000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x207000) = 0x76c2b1ebb000
mmap(0x76c2b1ebb000, 5648, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x76c2b1ebb000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832
read64(3, "\6\0\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 784, 64) = 784
read64(3, "\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 48, 848) = 48
read64(3, "\4\0\0\0\0\24\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 327, 321) = 327
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
read64(3, "\6\0\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 784, 64) = 784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76c2b1c00000
mprotect(0x76c2b1c28000, 2023424, PROT_NONE) = 0
mmap(0x76c2b1c28000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76c2b1c28000
mmap(0x76c2b1dbd000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x76c2b1dbd000
mmap(0x76c2b1dbd000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x76c2b1dbd000
mmap(0x76c2b1e1c000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x76c2b1e1c000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=613064, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 615184, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76c2b1b69000
mmap(0x76c2b1b6b000, 438272, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76c2b1b6b000
senjaramadhanil@LAPTOP-8230CJCF:~$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7fffd53aca60 /* 25 vars */) = 0
brk(NULL)
      = 0x5f1265582000
arch_prctl(0x3001 /* ARCH.??? */, 0x7ff99a7a29f0) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x79b361919000
access("/etc/ld.so.preload", R_OK)
      = -1 ENOENT (No such file or directory)
open(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=21996, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 21996, PROT_READ, MAP_PRIVATE, 3, 0) = 0x79b361913000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=166280, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 177672, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79b3618e7000
mprotect(0x79b3618ed000, 139264, PROT_NONE) = 0
mmap(0x79b3618ed000, 164996, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x6000) = 0x79b3618ed000
mmap(0x79b361907000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x79b361907000
mmap(0x79b36190f000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x207000) = 0x79b36190f000
mmap(0x79b361911000, 5640, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x79b361911000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832
read64(3, "\6\0\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 784, 64) = 784
read64(3, "\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 48, 848) = 48
read64(3, "\4\0\0\0\0\24\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 327, 321) = 327
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) = 0
read64(3, "\6\0\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0... 784, 64) = 784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x79b361600000
mprotect(0x79b361628000, 2023424, PROT_NONE) = 0
mmap(0x79b361628000, 1658880, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x79b361628000
mmap(0x79b3617bd000, 360448, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x79b3617bd000
mmap(0x79b361816000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x79b361816000
mmap(0x79b36181c000, 52816, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x79b36181c000
close(3)
      = 0
open(AT_FDCWD, "/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0\0\1\0\0\0\0\0\0\0\0\0\0\0\0... 832) = 832

```



```

senjaramadhanig@LAPTOP-823QCJCF:~$ ltrace ls
strchr("ls", '/')
= nil
setlocale(LC_ALL, "")
= "C.UTF-8"
bindtextdomain("coreutils", "/usr/share/locale")
= "/usr/share/locale"
textdomain("coreutils")
= "coreutils"
__cxa_atexit(0x60dea0d8da80, 0, 0x60dea0da4008, 0)
= 0
isatty(1)
= 1
getenv("QUOTING_STYLE")
= nil
getenv("COLUMNS")
= nil
ioctl(1, 21523, 0x7ffc1f184de0)
= 0
getenv("TABSIZ")
= nil
getopt_long(1, 0x7ffc1f184f48, "abcdfghiklmnopqrstuvw:ABCOFGHI:...", 0x60dea0da33a0, -1)
= -1
getenv("LS_BLOCK_SIZE")
= nil
getenv("BLOCK_SIZE")
= nil
getenv("BLOCKSIZE")
= nil
getenv("POSIXLY_CORRECT")
= nil
getenv("BLOCK_SIZE")
= nil
__errno_location()
= 0x7cd2986f9698
__errno_location()
= 0x7cd2986f9698
getenv("TZ")
= nil
strlen(".")
= 1
memcpy(0x60deb969f560, ".\0", 2)
= 0x60deb969f560
__errno_location()
= 0x7cd2986f9698
opendir(".")
= 0x60deb969f580
readdir(0x60deb969f580)
= 0x60deb969f5b0
readdir(0x60deb969f580)
= 0x60deb969f5e0
__errno_location()
= 0x7cd2986f9698
__ctype_get_mb_cur_max()
= 6
strlen("math_ops.o")
= 10
strlen("math_ops.o")
= 10
memcpy(0x60deb96a75c0, "math_ops.o\0", 11)
= 0x60deb96a75c0
readdir(0x60deb969f580)
= 0x60deb969f600
__errno_location()
= 0x7cd2986f9698
__ctype_get_mb_cur_max()
= 6
fileno(0x7cd29861b6a0)
= 2
__freanding(0x7cd29861b6a0, 0, 0x7cd298616a00, 3072)
= 0
__freanding(0x7cd29861b6a0, 0, 0x7cd298616a00, 3072)
= 0
fflush(0x7cd29861b6a0)
= 0
fclose(0x7cd29861b6a0)
= 0
+++ exited (status 0) +++
senjaramadhanig@LAPTOP-823QCJCF:~$ ltrace -c ls
Makefile      fork_demo.c   ipc_benchmark.c  math_ops.c     my_daemon.c     process_basic.c  program_dynamic  shm_writer.c    writer
WSL2-Linux-Kernel  fork_demo.c   libmathops.so    math_ops.h     my_daemon_dir   process_syscalls  program_shared   shmfile
bootloader_concept.asm  hello_mod     main.c           math_ops.i     my_daemon_log.txt  process_syscalls.c  program_static   simple_shell
fifo_reader.c         hello_module.c  main.i           math_ops.o     os_services.c     process_wait.c     reader           simple_shell.c
fifo_writer.c         hello_module.ko  main.o           math_ops.s     os_specific.c     procfs_read.c     shm_reader       syscall_test.txt
file_syscalls         info_syscalls.c  main.s           multiple_fork.c  pipe_demo.c       procfs_read.c     shm_reader.c     test_dir
file_syscalls.c       info_syscalls.c  makefile         my_daemon      process_basic      program           shm_writer       trace.txt
% time    seconds  usecs/call      calls      function
-----
20.27    0.060920      189           322    __errno_location
17.50    0.052597      184           285    __ctype_get_mb_cur_max
16.92    0.050860      177           286    strlen
16.01    0.048124      183           262    strcoll
11.34    0.034091      195           174    __overflow
5.25     0.015783      185            85    memcpy
4.02     0.012077      170            71    readdir
3.56     0.010686      187            57    fwrite_unlocked
2.87     0.008638      8638            1    setlocale
0.44     0.001316      146             9    getenv
0.34     0.001009      504             2    fclose
0.21     0.000620      155             4    __freanding
0.15     0.000446      446             1    opendir
0.12     0.000368      368             1    textdomain
0.11     0.000344      172             2    fflush
0.11     0.000316      158             2    __pending
0.10     0.000315      157             2    fileno
0.10     0.000296      206             1    closedir
0.09     0.000276      276             1    isatty
0.08     0.000254      254             1    __cxa_atexit
0.08     0.000246      246             1    ioctl
0.08     0.000227      227             1    bindtextdomain
0.08     0.000226      226             1    strchr
0.06     0.000187      187             1    realloc
0.05     0.000152      152             1    setjmp
0.05     0.000149      149             1    getopt_long
-----
100.00    0.300523                1575    total

```

Praktik 10.3: Using GDB

Buat program dengan bug buggy.c:

```
senjaramadhani@LAPTOP-823QCJCF:~$ nano buggy.c
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -g -o buggy buggy.c
senjaramadhani@LAPTOP-823QCJCF:~$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
senjaramadhani@LAPTOP-823QCJCF:~$ sudo apt install gdb
[sudo] password for senjaramadhani:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libbabeltrace1 libboost-regex1.74.0 libc6-dbg libdebuginfod-common libdebuginfod1 libipt2 libsource-highlight-common
  libsource-highlight4v5
Suggested packages:
  gdb-doc gdbserver
The following NEW packages will be installed:
  gdb libbabeltrace1 libboost-regex1.74.0 libc6-dbg libdebuginfod-common libdebuginfod1 libipt2
  libsource-highlight-common libsource-highlight4v5
0 upgraded, 9 newly installed, 0 to remove and 40 not upgraded.
Need to get 13.9 MB/18.8 MB of archives.
After this operation, 35.1 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-dbg amd64 2.35-0ubuntu3.11 [13.9 MB]
Fetched 2329 kB in 4s (562 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libdebuginfod-common.
(Reading database ... 51575 files and directories currently installed.)
Preparing to unpack .../0-libdebuginfod-common_0.186-1ubuntu0.1_all.deb ...
Unpacking libdebuginfod-common (0.186-1ubuntu0.1) ...
Selecting previously unselected package libbabeltrace1:amd64.
Preparing to unpack .../1-libbabeltrace1_1.5.8-2build1_amd64.deb ...
Unpacking libbabeltrace1:amd64 (1.5.8-2build1) ...
Selecting previously unselected package libdebuginfod1:amd64.
Preparing to unpack .../2-libdebuginfod1_0.186-1ubuntu0.1_amd64.deb ...
Unpacking libdebuginfod1:amd64 (0.186-1ubuntu0.1) ...
Selecting previously unselected package libipt2.
Preparing to unpack .../3-libipt2_2.0.5-1_amd64.deb ...
Unpacking libipt2 (2.0.5-1) ...
Selecting previously unselected package libsource-highlight-common.
Preparing to unpack .../4-libsource-highlight-common_3.1.9-4.1build2_all.deb ...
senjaramadhani@LAPTOP-823QCJCF:~$ gdb ./buggy
GNU gdb (Ubuntu 12.1-0ubuntu1~22.04.2) 12.1
Copyright (C) 2022 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
Type "show copying" and "show warranty" for details.
This GDB was configured as "x86_64-linux-gnu".
Type "show configuration" for configuration details.
For bug reporting instructions, please see:
<https://www.gnu.org/software/gdb/bugs/>.
Find the GDB manual and other documentation resources online at:
<http://www.gnu.org/software/gdb/documentation/>.

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...

For help, type "help".
Type "apropos word" to search for commands related to "word"...
Reading symbols from ./buggy...
(gdb) break main
Breakpoint 1 at 0x1238: file buggy.c, line 17.
(gdb) run
Starting program: /home/senjaramadhani/buggy

[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at buggy.c:17
17      buggy_function(5);
(gdb)
(gdb) next
array[0] = 0
array[1] = 2
array[2] = 4
array[3] = 6
array[4] = 8
18      return 0;
(gdb) step
19      }
(gdb) print n
No symbol "n" in current context.
(gdb) continue
Continuing.
[Inferior 1 (process 774) exited normally]
(gdb) backtrace
No stack.
(gdb) quit
senjaramadhani@LAPTOP-823QCJCF:~$ |
```

Praktik 10.4: Kernel Debugging dengan dmesg

```
senjaramadhani@LAPTOP-823QCJCF:~$ dmesg
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu
Jun 5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_E
NABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000a0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001023ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[ 0.000000] Hyper-V: Nested features: 0x4a0000
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 3094.109 MHz processor
[ 0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.000000] last_pfn = 0x102400 max_arch_pfn = 0x40000000
[ 0.000113] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.000115] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[ 0.000154] last_pfn = 0xf8000 max_arch_pfn = 0x40000000
[ 0.000165] Using GB pages for direct mapping
[ 0.000253] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[ 0.000255] ACPI: Early table checksum verification disabled
[ 0.000258] ACPI: RSDP 0x00000000000E0000 000024 (v02 VIRTUAL)
[ 0.000261] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)

senjaramadhani@LAPTOP-823QCJCF:~$ dmesg -w
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu
Jun 5 18:30:46 UTC 2025
[ 0.000000] Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_E
NABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000a0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001023ffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMI not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0x900c2c, misc 0xe0bed7b6
[ 0.000000] Hyper-V: Nested features: 0x4a0000
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
[ 0.000000] Hyper-V: Using hypercall for remote TLB flush
[ 0.000000] clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
[ 0.000000] tsc: Detected 3094.109 MHz processor
[ 0.000000] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.000000] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.000000] last_pfn = 0x102400 max_arch_pfn = 0x40000000
[ 0.000113] MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
[ 0.000115] x86/PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
[ 0.000154] last_pfn = 0xf8000 max_arch_pfn = 0x40000000
[ 0.000165] Using GB pages for direct mapping
[ 0.000253] RAMDISK: [mem 0x04be4000-0x04e7ffff]
[ 0.000255] ACPI: Early table checksum verification disabled
[ 0.000258] ACPI: RSDP 0x00000000000E0000 000024 (v02 VIRTUAL)
[ 0.000261] ACPI: XSDT 0x0000000000100000 000044 (v01 VIRTUAL MICROSOFT 00000001 MSFT 00000001)

senjaramadhani@LAPTOP-823QCJCF:~$ dmesg -l err
[ 0.145778] PCI: Fatal: No config space access function found
[ 2.258975] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 2.262458] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 2.263079] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 2.263673] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 2.264175] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 2.707405] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.274204] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.275440] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.276638] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.277840] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.278939] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.280160] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.281380] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.282733] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.283670] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.285370] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.286370] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 3.494655] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 6.210349] WSL (261) ERROR: CheckConnection: getaddrinfo() failed: -5
senjaramadhani@LAPTOP-823QCJCF:~$ dmesg -l warn
[ 0.120198] Speculative Return Stack Overflow: IBPB-extending microcode not applied!
[ 0.120198] Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
[ 0.181235] PCI: System does not support PCI
[ 0.336227] device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
[ 2.118836] pulseaudio[296]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 2.373994] Failed to connect to bus: No such file or directory
[ 2.625848] systemd-journal[61]: File /var/log/journal/4d24d7d873da4319ae2ce7ab31678dea/system.journal corrupted or uncleanly shut down, renaming and re
placing.
[ 4.540179] systemd-journal[61]: File /var/log/journal/4d24d7d873da4319ae2ce7ab31678dea/user-1000.journal corrupted or uncleanly shut down, renaming and
replacing.
[ 4.876227] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
senjaramadhani@LAPTOP-823QCJCF:~$
```

Praktik 10.5: Memory Leak Detection dengan Valgrind

```
senjaramadhani@LAPTOP-823QCJCF:~$ sudo apt install valgrind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libc6-i386
Suggested packages:
  valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  libc6-i386 valgrind
0 upgraded, 2 newly installed, 0 to remove and 40 not upgraded.
Need to get 16.9 MB of archives.
After this operation, 91.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-i386 amd64 2.35-0ubuntu3.11 [2838 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 valgrind amd64 1:3.18.1-1ubuntu2 [14.1 MB]
Fetched 16.9 MB in 7s (2443 kB/s)
Selecting previously unselected package libc6-i386.
(Reading database ... 52378 files and directories currently installed.)
Preparing to unpack .../libc6-i386_2.35-0ubuntu3.11_amd64.deb ...
Unpacking libc6-i386 (2.35-0ubuntu3.11) ...
Selecting previously unselected package valgrind.
Preparing to unpack .../valgrind_1%3a3.18.1-1ubuntu2_amd64.deb ...
Unpacking valgrind (1:3.18.1-1ubuntu2) ...
Setting up libc6-i386 (2.35-0ubuntu3.11) ...
Setting up valgrind (1:3.18.1-1ubuntu2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.11) ...
senjaramadhani@LAPTOP-823QCJCF:~$ cat > memleak.c << 'EOF'
> <stdlib.h>
> ^C
senjaramadhani@LAPTOP-823QCJCF:~$ cat > memleak.c << 'EOF'
> int main() {
  int *ptr = (int*)malloc(100 * sizeof(int));
  // Forgot to free(ptr)
  return 0;
}
EOF
senjaramadhani@LAPTOP-823QCJCF:~$ gcc -g -o memleak memleak.c
memleak.c: In function 'main':
memleak.c:2:19: warning: implicit declaration of function 'malloc' [-Wimplicit-function-declaration]
   2 |   int *ptr = (int*)malloc(100 * sizeof(int));
     |                     ^~~~~~
memleak.c:1:1: note: include '<stdlib.h>' or provide a declaration of 'malloc'
+++ |#include <stdlib.h>
   1 | int main() {
memleak.c:2:19: warning: incompatible implicit declaration of built-in function 'malloc' [-Wbuiltin-declaration-mismatch]
   2 |   int *ptr = (int*)malloc(100 * sizeof(int));
     |                     ^~~~~~
memleak.c:2:19: note: include '<stdlib.h>' or provide a declaration of 'malloc'
senjaramadhani@LAPTOP-823QCJCF:~$ valgrind --leak-check=full ./memleak
==912== Memcheck, a memory error detector
==912== Copyright (C) 2002-2017, and GNU GPL'd, by Julian Seward et al.
==912== Using Valgrind-3.18.1 and LibVEX; rerun with -h for copyright info
==912== Command: ./memleak
==912==
==912== HEAP SUMMARY:
==912==   in use at exit: 400 bytes in 1 blocks
==912==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==912==
==912== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==912==   at 0x4848899: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==912==   by 0x10915E: main (memleak.c:2)
==912==
==912== LEAK SUMMARY:
==912==   definitely lost: 400 bytes in 1 blocks
==912==   indirectly lost: 0 bytes in 0 blocks
==912==   possibly lost: 0 bytes in 0 blocks
==912==   still reachable: 0 bytes in 0 blocks
==912==   suppressed: 0 bytes in 0 blocks
==912==
==912== For lists of detected and suppressed errors, rerun with: -s
==912== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
senjaramadhani@LAPTOP-823QCJCF:~$
```

Praktik 10.6: Performance Analysis

```
senjaramadhani@LAPTOP-823QCJCF:~$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  hwdata linux-tools-5.15.0-157 linux-tools-5.15.0-157-generic
The following NEW packages will be installed:
  hwdata linux-tools-5.15.0-157 linux-tools-5.15.0-157-generic linux-tools-common linux-tools-generic
0 upgraded, 5 newly installed, 0 to remove and 40 not upgraded.
Need to get 8279 kB of archives.
After this operation, 29.3 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 hwdata all 0.357-1 [26.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-common all 5.15.0-157.167 [301 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-5.15.0-157 amd64 5.15.0-157.167 [7947 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-5.15.0-157-generic amd64 5.15.0-157.167 [1792 B]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-tools-generic amd64 5.15.0-157.155 [2334 B]
Fetched 8279 kB in 10s (829 kB/s)
Selecting previously unselected package hwdata.
(Reading database ... 53833 files and directories currently installed.)
Preparing to unpack .../hwdata_0.357-1_all.deb ...
Unpacking hwdata (0.357-1) ...
Selecting previously unselected package linux-tools-common.
Preparing to unpack .../linux-tools-common_5.15.0-157.167_all.deb ...
Unpacking linux-tools-common (5.15.0-157.167) ...
Selecting previously unselected package linux-tools-5.15.0-157.
Preparing to unpack .../linux-tools-5.15.0-157_5.15.0-157.167_amd64.deb ...
Unpacking linux-tools-5.15.0-157 (5.15.0-157.167) ...
Selecting previously unselected package linux-tools-5.15.0-157-generic.
Preparing to unpack .../linux-tools-5.15.0-157-generic_5.15.0-157.167_amd64.deb ...
Unpacking linux-tools-5.15.0-157-generic (5.15.0-157.167) ...
Selecting previously unselected package linux-tools-generic.
Preparing to unpack .../linux-tools-generic_5.15.0-157.155_amd64.deb ...
Unpacking linux-tools-generic (5.15.0-157.155) ...
Setting up hwdata (0.357-1) ...
Setting up linux-tools-common (5.15.0-157.167) ...
Setting up linux-tools-5.15.0-157 (5.15.0-157.167) ...
Setting up linux-tools-5.15.0-157-generic (5.15.0-157.167) ...
Setting up linux-tools-generic (5.15.0-157.155) ...
Processing triggers for man-db (2.10.2-1) ...
senjaramadhani@LAPTOP-823QCJCF:~$ sudo perf trace ls
```